

Convergence Tests

ANSWER KEY



The ratio and comparison tests

- 1 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
 $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2} < 1$: converges.
- 2 Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.
 $\frac{a_{n+1}}{a_n} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \rightarrow \frac{1}{e} < 1$: converges.
- 3 Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum_{n=1}^{\infty} \frac{1}{n^2}$.
 $0 < \frac{1}{n^2+3} < \frac{1}{n^2}$ for all $n \geq 1$, and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent p -series ($p = 2 > 1$), so by comparison the given series converges.

Alternating series and the p-series test

- 4 Determine whether each p -series converges or diverges:
a $\sum_{n=1}^{\infty} \frac{1}{n^3}$ Converges ($p = 3 > 1$). **b** $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ Diverges ($p = \frac{1}{2} \leq 1$).
- 5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.
The terms $\frac{1}{n}$ decrease to 0, so by the alternating series test the series converges. But $\sum_{n=1}^{\infty} \frac{1}{n}$ (absolute values) diverges — so the series is conditionally convergent.

The integral test

- 6 Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_2^{\infty} \frac{1}{x \ln x} dx$.
Let $u = \ln x$: $\int_2^{\infty} \frac{1}{x \ln x} dx = \ln|\ln x| + C$. $\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} [\ln(\ln x)]_2^b = \infty$: the integral diverges, so by the integral test the series diverges.
- 7 Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_1^{\infty} \frac{1}{x^2} dx$.
 $\int_1^{\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{x}\right]_1^b = 1$, a finite value, so by the integral test the series converges.

Choosing the right test

- 8 For each series, name the most efficient test to apply (ratio, comparison, p -series, or alternating series), without fully evaluating:
- a $\sum \frac{2^n}{n!}$ Ratio test (factorial in denominator is a strong signal).
 - b $\sum \frac{(-1)^n}{\sqrt{n}}$ Alternating series test (terms decrease to 0).
 - c $\sum \frac{1}{n^4}$ p -series test directly ($p = 4 > 1$: converges).
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The limit comparison test

- 9 Use the limit comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$ converges, comparing to $\sum \frac{1}{n^2}$.
 $\lim_{n \rightarrow \infty} \frac{(n+1)/(n^3+2)}{1/n^2} = \lim_{n \rightarrow \infty} \frac{n^3+n^2}{n^3+2} = 1$, a finite positive value. Since $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.
- 10 Use the limit comparison test on $\sum_{n=1}^{\infty} \frac{3n^2-1}{n^4+2n}$, comparing to $\sum \frac{1}{n^2}$.
 $\lim_{n \rightarrow \infty} \frac{(3n^2-1)/(n^4+2n)}{1/n^2} = \lim_{n \rightarrow \infty} \frac{3n^4-n^2}{n^4+2n} = 3$, finite and positive. Since $\sum \frac{1}{n^2}$ converges, so does the given series.
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The root test

- 11 Use the root test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.
 $\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{n}{2n+1}\right)^n} = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2} < 1$: converges.
- 12 Use the root test on $\sum_{n=1}^{\infty} \frac{3^n}{n^n}$.
 $\lim_{n \rightarrow \infty} \sqrt[n]{\frac{3^n}{n^n}} = \lim_{n \rightarrow \infty} \frac{3}{n} = 0 < 1$: converges.
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Absolute versus conditional convergence

- 13 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent, conditionally convergent, or divergent.
 Taking absolute values: $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent p -series ($p = 2 > 1$). Since the series of absolute values converges, the original series is absolutely convergent.
- 14 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ is absolutely convergent, conditionally convergent, or divergent.
 Absolute values give $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$, a divergent p -series ($p = \frac{1}{2} \leq 1$). But the terms $\frac{1}{\sqrt{n}}$ decrease to 0, so by the alternating series test the original series converges. It is conditionally convergent.
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Estimating the sum of an alternating series

- 15 The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges by the alternating series test. Use the fact that the error after n terms is bounded by the first omitted term to estimate how many terms are needed for the partial sum to be within 0.001 of the true sum.
- Error bound: $|S - S_n| \leq a_{n+1} = \frac{1}{(n+1)^2}$. Need $\frac{1}{(n+1)^2} < 0.001$: $(n+1)^2 > 1000$, so $n+1 > 31.6$, meaning $n \geq 32$ terms suffice.

A combined convergence-test challenge

- 16 Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ converges, using the ratio test.
- $\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^2}{3^{n+1}} \cdot \frac{3^n}{n^2} = \frac{1}{3} \left(\frac{n+1}{n} \right)^2 \rightarrow \frac{1}{3} < 1$: converges.
- 17 Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the integral test with $u = \ln x$.
- $\int_2^{\infty} \frac{1}{x(\ln x)^2} dx$: with $u = \ln x$, $du = \frac{1}{x} dx$: $\int \frac{1}{u^2} du = -\frac{1}{u} + C$. Evaluating: $\lim_{b \rightarrow \infty} \left[-\frac{1}{\ln x} \right]_2^b = 0 - \left(-\frac{1}{\ln 2} \right) = \frac{1}{\ln 2}$, a finite value: the integral converges, so by the integral test the series converges.
- 18 A student claims that since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ must converge. Explain what is wrong with this reasoning, and state the correct necessary (but not sufficient) condition that the n th-term test actually provides, using the harmonic series as a counterexample.
- The reasoning confuses a necessary condition with a sufficient one. The n th-term test only says: IF $\lim_{n \rightarrow \infty} a_n \neq 0$, THEN the series diverges. It says nothing when the limit IS 0 — that case is inconclusive. The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the standard counterexample: its terms go to 0, yet the series diverges (proven via the integral test or the comparison/grouping argument), showing that $\lim_{n \rightarrow \infty} a_n = 0$ alone never guarantees convergence.

A multi-step convergence investigation

- 19 For the series $\sum_{n=1}^{\infty} \frac{n^3 + 2n}{2n^5 - n + 1}$, first find the n th-term limit to check the necessary condition, then use the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^2}$ to determine whether the series converges.
- n th-term limit: $\lim_{n \rightarrow \infty} \frac{n^3 + 2n}{2n^5 - n + 1} = 0$ (inconclusive by itself). Limit comparison with $\frac{1}{n^2}$:
- $\lim_{n \rightarrow \infty} \frac{(n^3 + 2n)/(2n^5 - n + 1)}{1/n^2} = \lim_{n \rightarrow \infty} \frac{n^5 + 2n^3}{2n^5 - n + 1} = \frac{1}{2}$, finite and positive. Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.
- 20 A quality-control engineer models cumulative defect probability using $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$. Use the ratio test to confirm this series converges, and briefly explain (in one sentence) why a convergent bound like this is useful for guaranteeing that a manufacturing process's total defect risk stays finite over infinitely many production runs.
- $\left| \frac{a_{n+1}}{a_n} \right| = \frac{n \cdot 2^n}{(n+1) \cdot 2^{n+1}} = \frac{n}{2(n+1)} \rightarrow \frac{1}{2} < 1$: converges. Practically, a convergent bound guarantees the cumulative defect probability approaches a finite ceiling rather than growing without bound as more production runs occur, which is essential for meaningful long-run risk assessment.